KO: Theoretical Test 1

March 16, 2021

1 Rules

- Solve two problems given on the following pages by yourself.
- Use a blank sheet of paper for your answers and do not forget to put your name on the top corner.
- You can ask for clarifications regarding the problem statement via Teams in the designated channel.
- We would like to remind you that you are not allowed to use any materials or the Internet.
- Scan/take a photo of your solution and submit it via BRUTE.

School lunch menu design

Let $M = \{1, \ldots, 100\}$ be a set of different meal types (e.g. schnitzel, lasagne etc.) and let $D = \{1, \ldots, 5\}$ be working days of one specific week. Each portion of meal $m \in M$ provides p_m proteins. For each meal $m \in M$, value $v_m \in \{0, 1\}$ determines whether it is vegetarian or not and value c_m is a cost of cooking one portion of meal m.

The goal is to design a school lunch menu for one week such that

1. each day, four different meals from M are on the menu and at least one of them is vegetarian.
2. the cost of the week menu has to be minimized. Every day, exactly 120 portions of each meal on the menu is cooked.
3. each meal $m \in M$ is on the menu at most once per week.
4. to satisfy recommended weekly limits on the nutrition values, it must hold that if any student orders arbitrary combination of meals in the week (exactly one meal per day), his/her total intake of protein must be at least $p^{\min}$.

Formulate this problem as **Integer Linear Programming**.

Branch and Bound Method for ILP

Solve the following ILP instance by Branch and Bound Algorithm, where each vertex of the search tree represents one LP problem (use graphical method to find the solution of such LP).

$$
\begin{aligned}
\min z = \quad & -x_1 & + 2x_2 & \\
s.t. \quad & 2x_1 & + \ x_2 & \geq 4 \\
& -4x_1 & + 4x_2 & \geq -5 \\
& & x_1, x_2 & \in \mathbb{Z}_0^+
\end{aligned}
$$